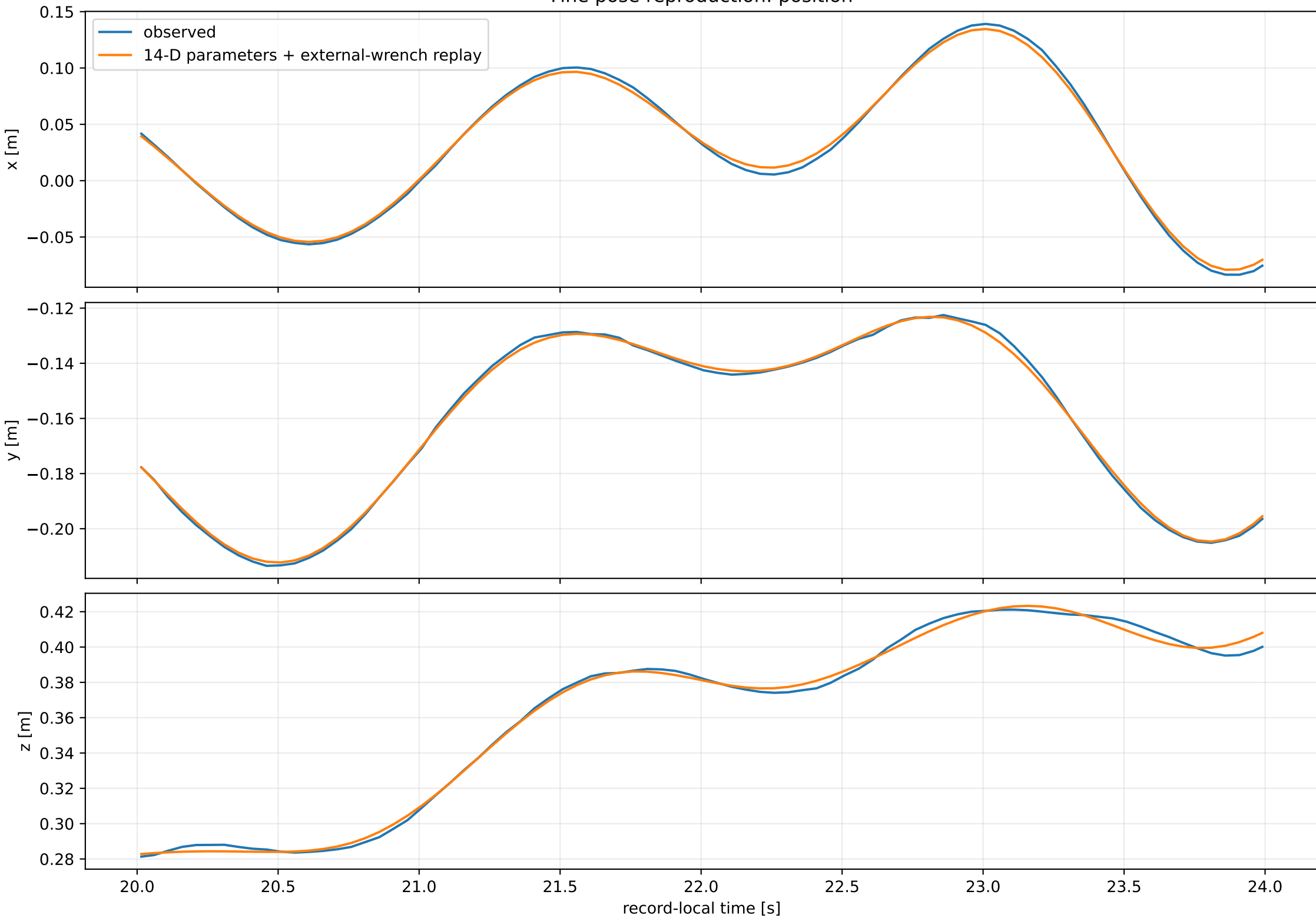
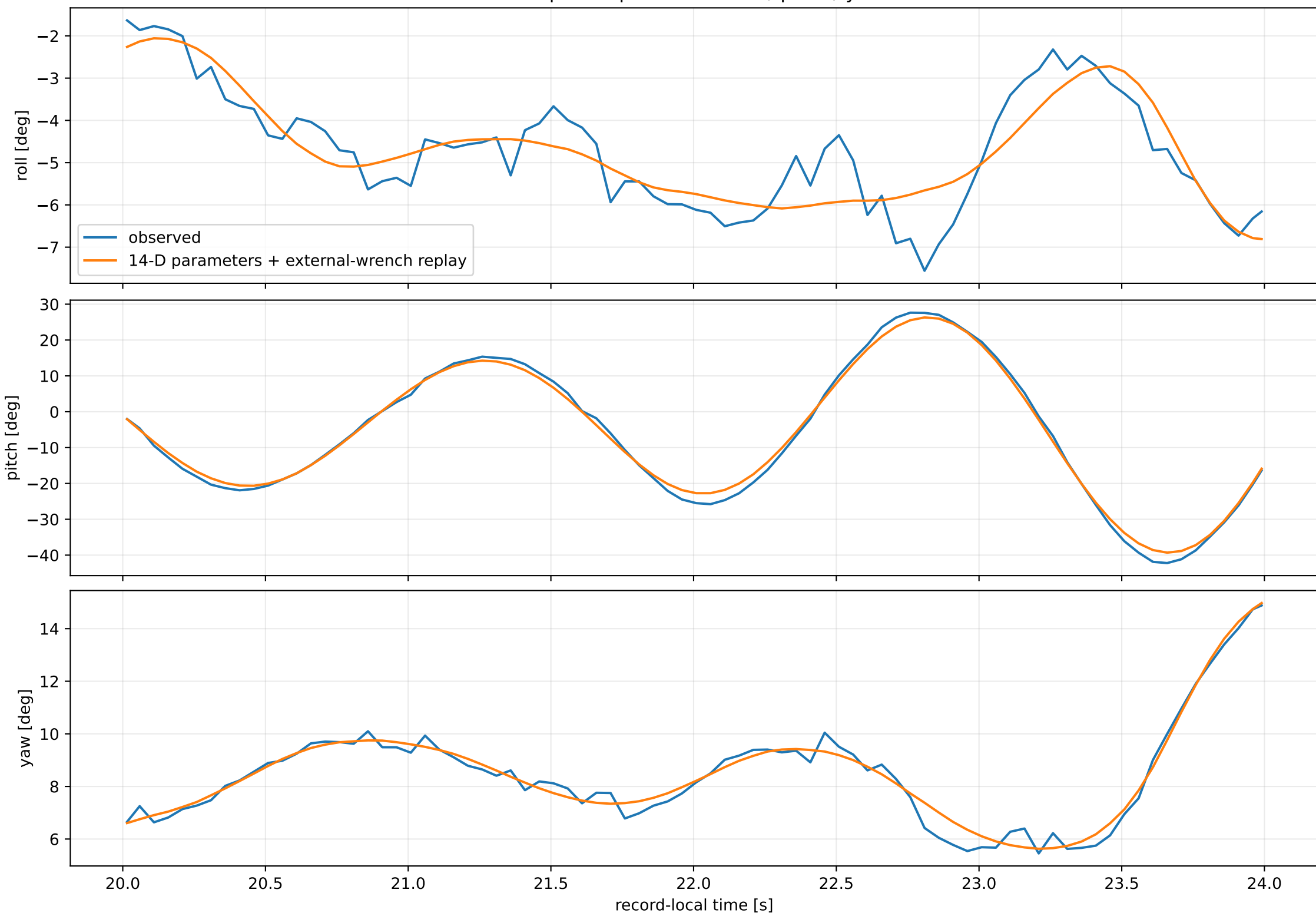


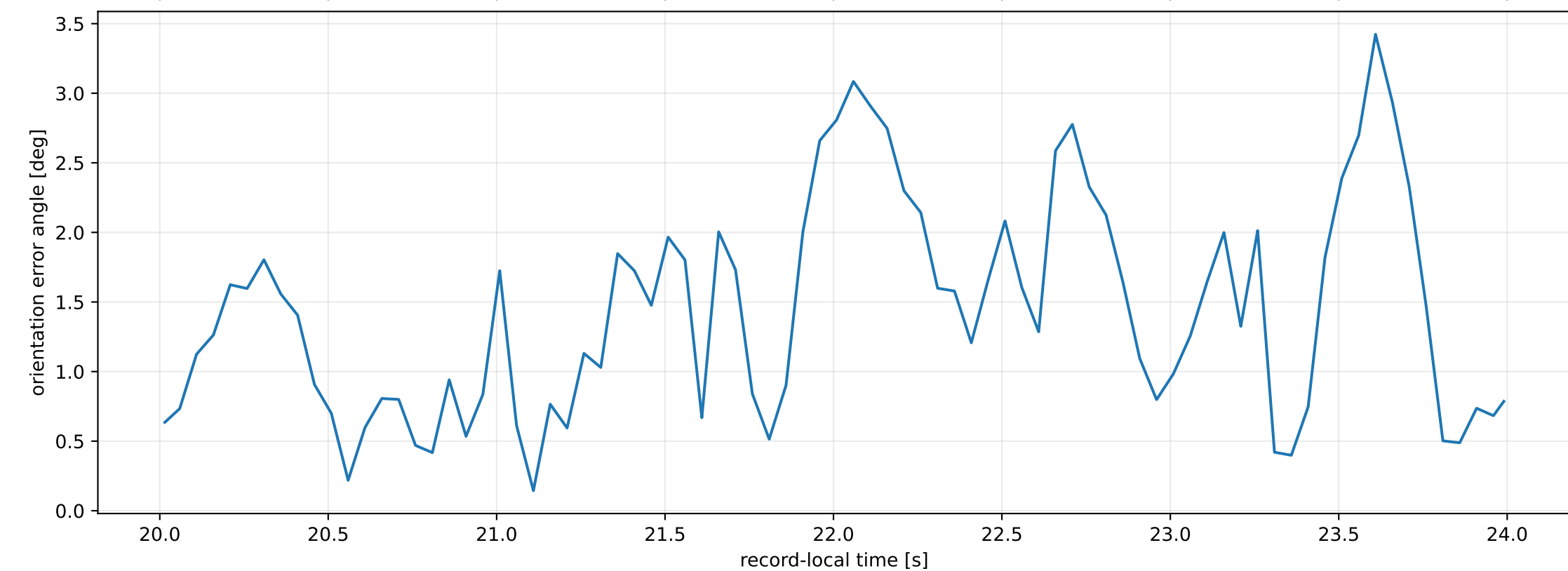
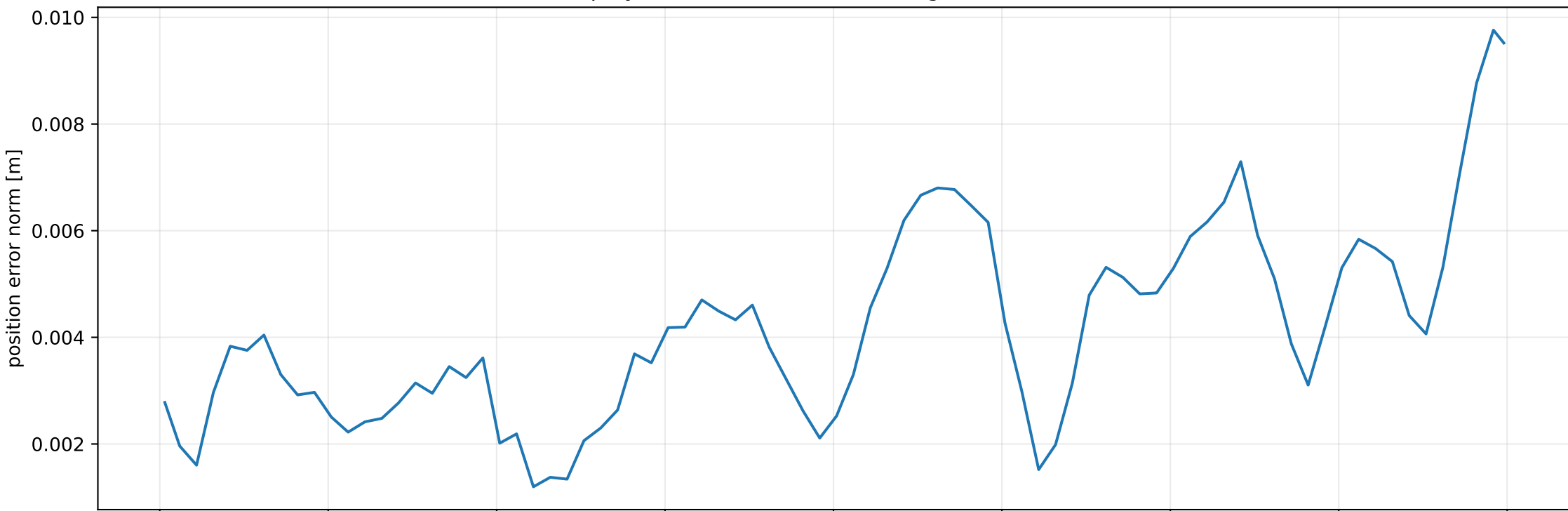
Fine pose reproduction: position



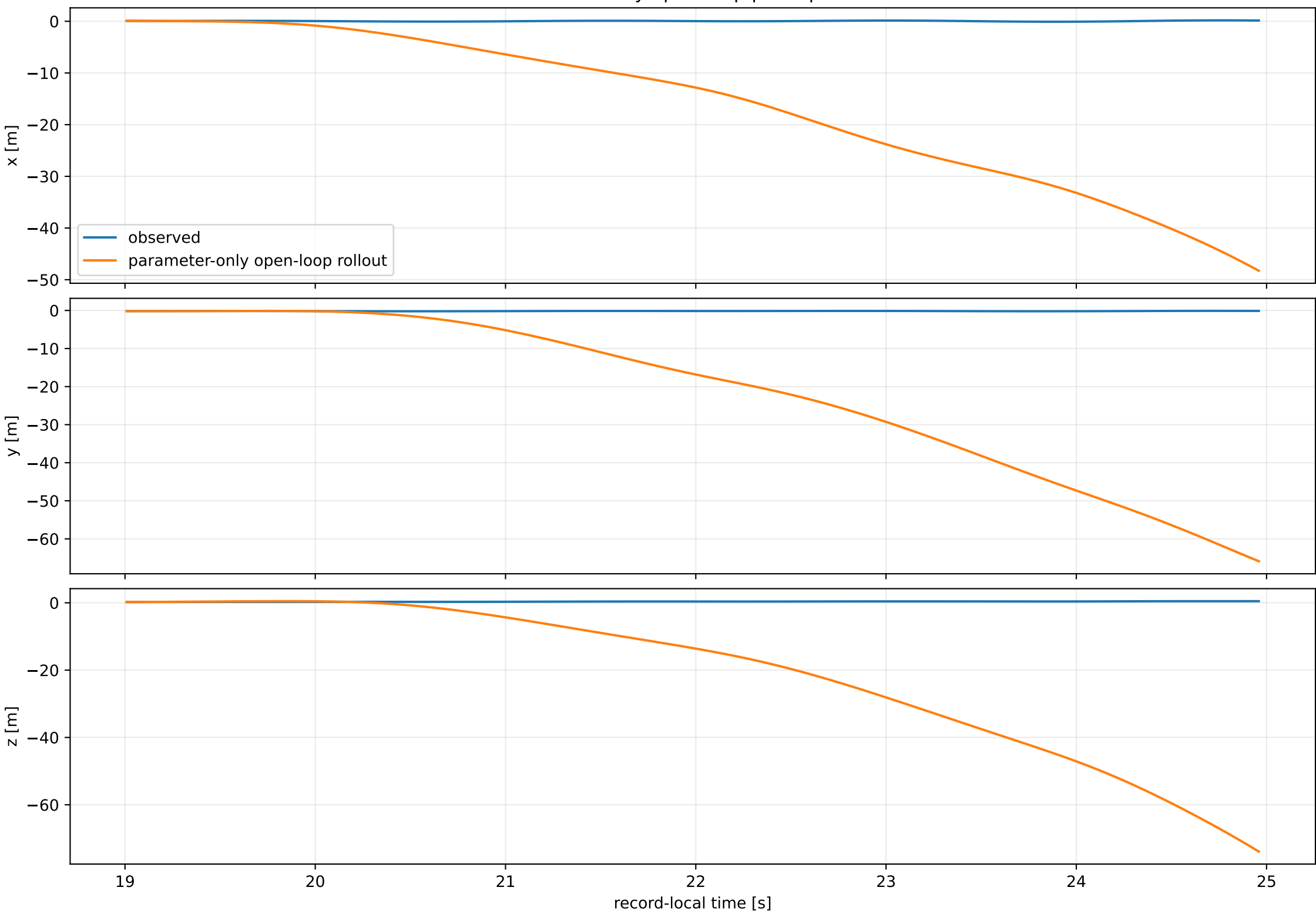
Fine pose reproduction: roll / pitch / yaw



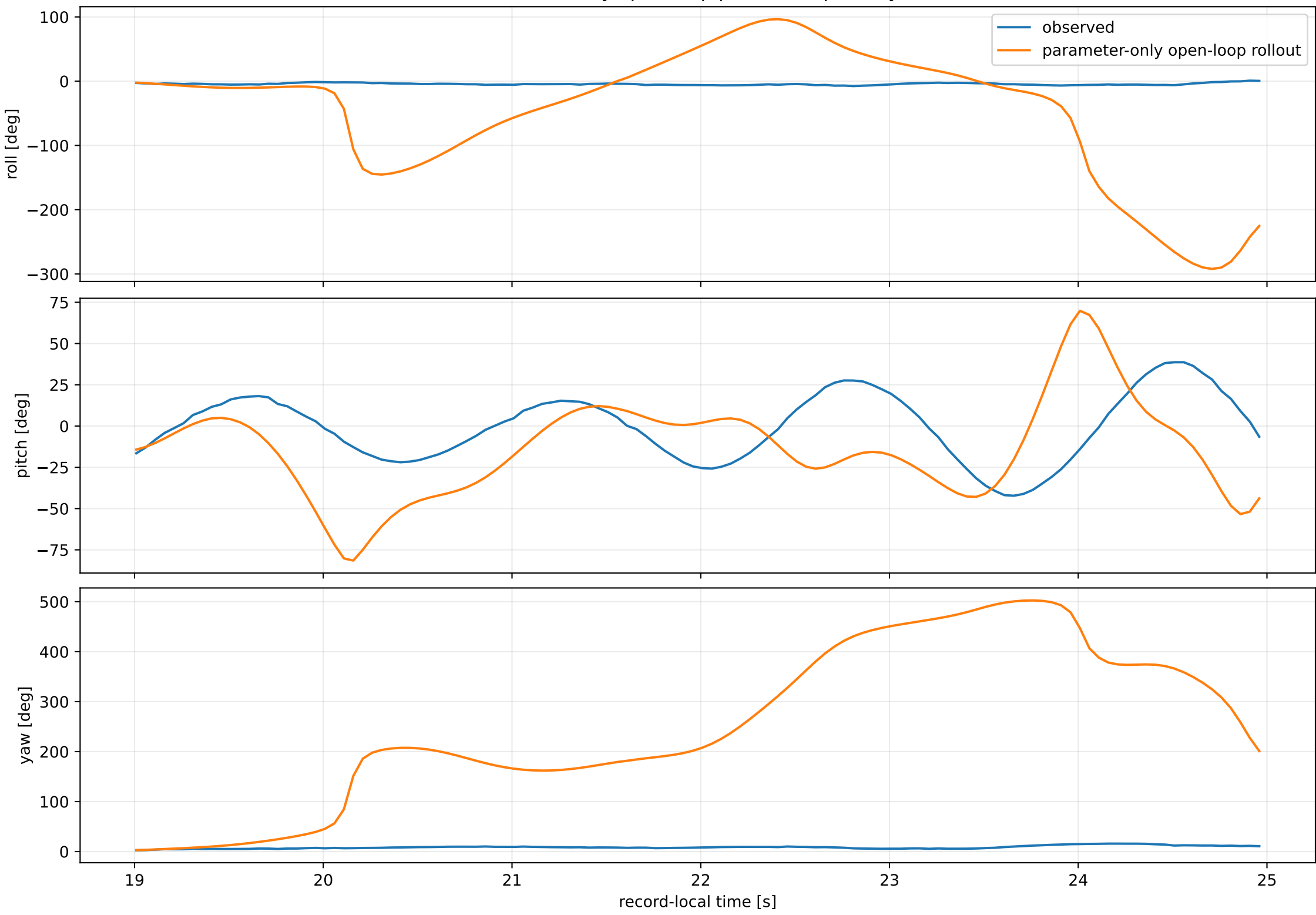
Fine replay error after the 14-D change — lower is better



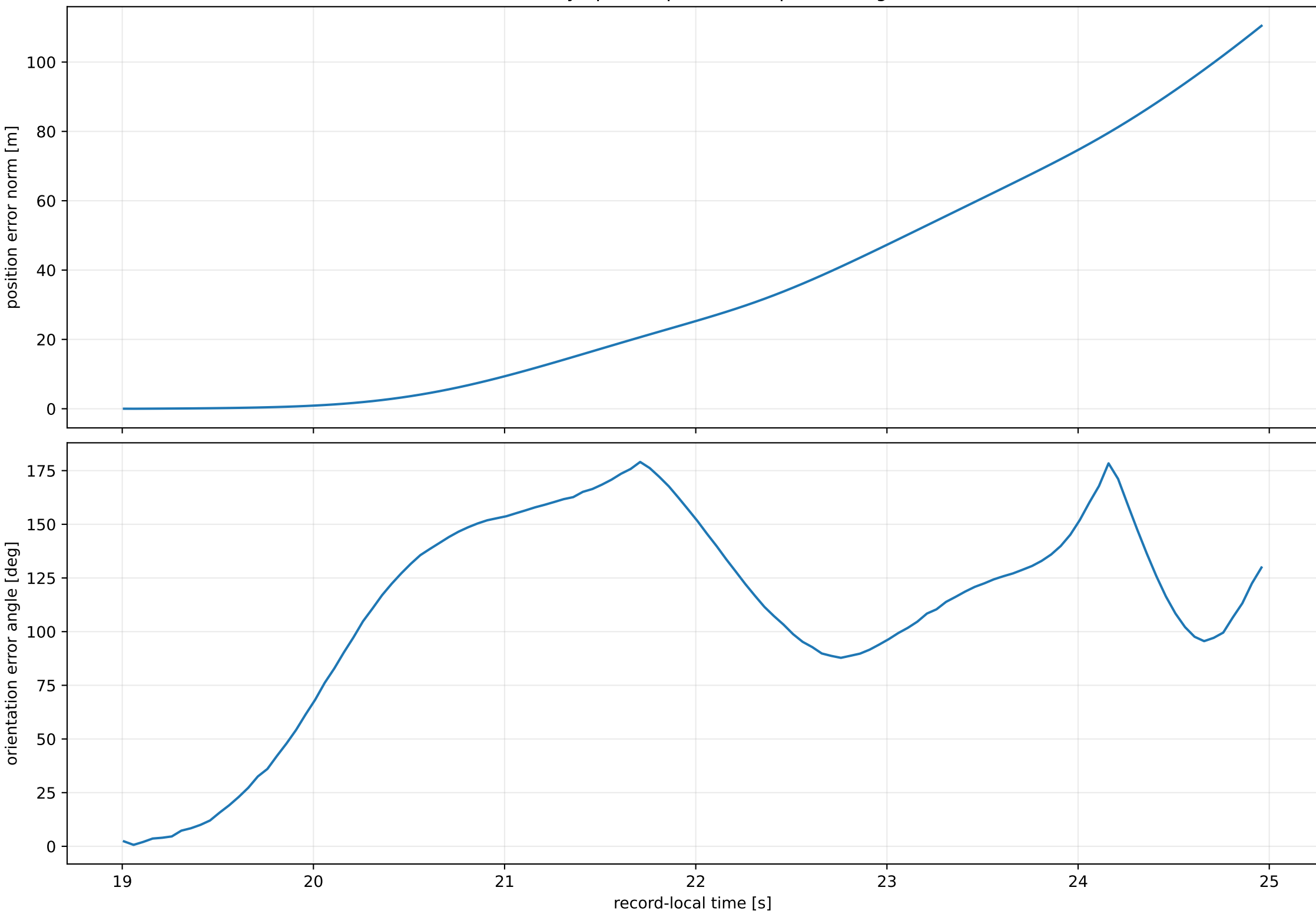
Parameter-only open-loop pose: position



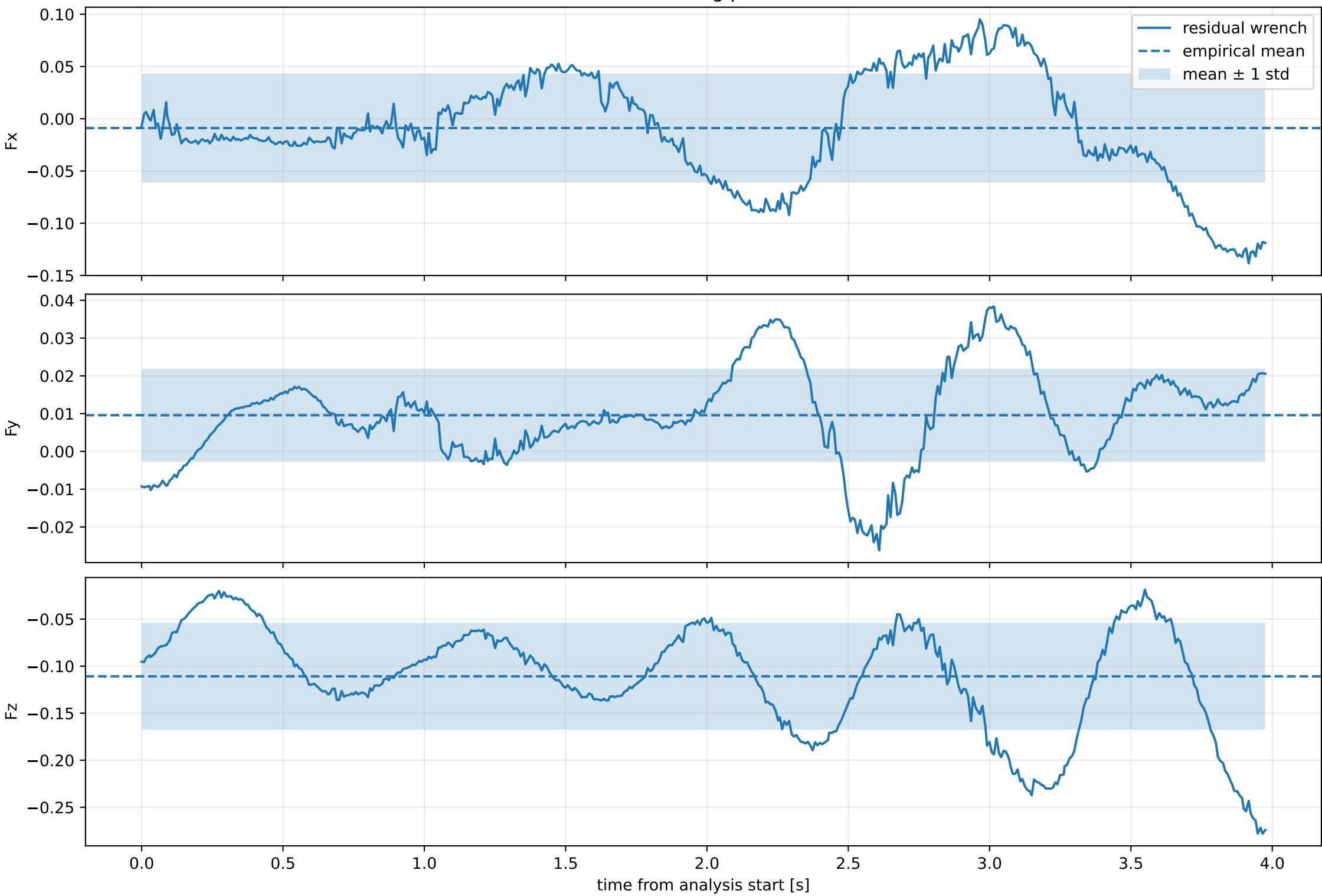
Parameter-only open-loop pose: roll / pitch / yaw



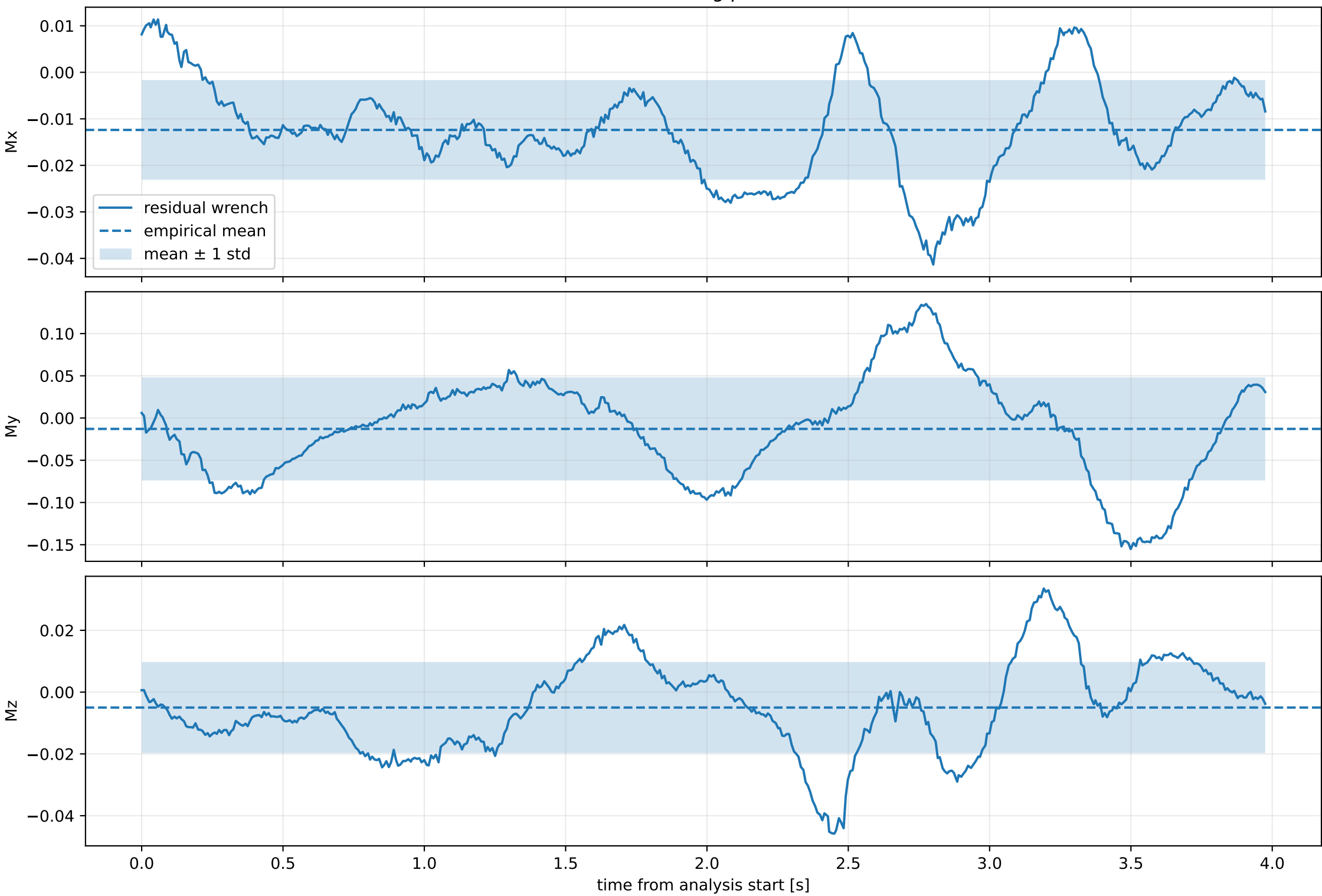
Parameter-only open-loop rollout (separate diagnostic)



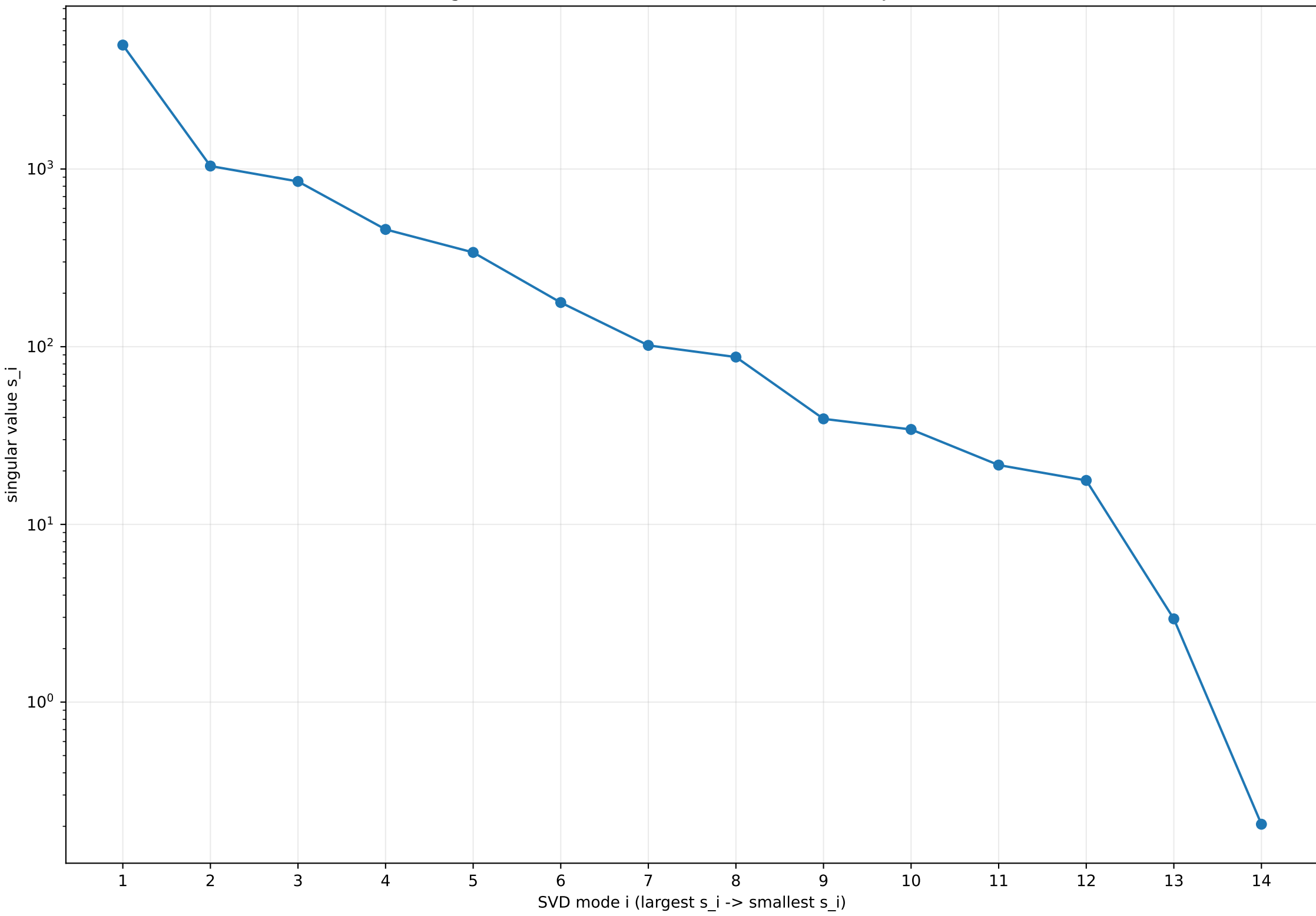
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



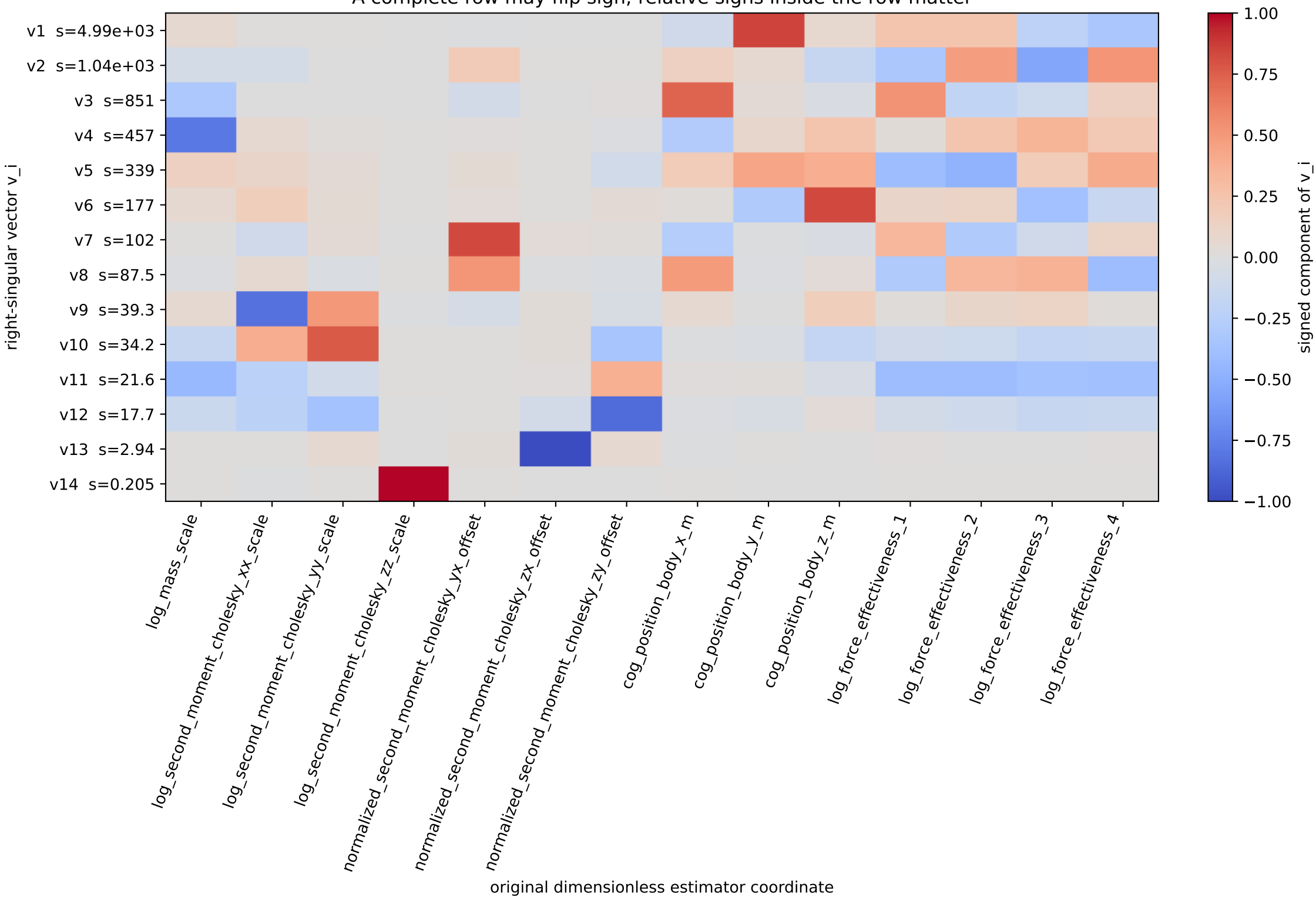
Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36566714716

inertia J [kg m²]

[0.065013625 -2.2810531e-07 2.409352e-05]
[-2.2810531e-07 0.065294473 1.194197e-05]
[2.409352e-05 1.194197e-05 0.12899456]

principal moments [kg m²]

[0.065013616, 0.065294471, 0.12899457]

CoG position in body frame [m]

[-0.0028310769, 0.00085087295, 0.010886177]

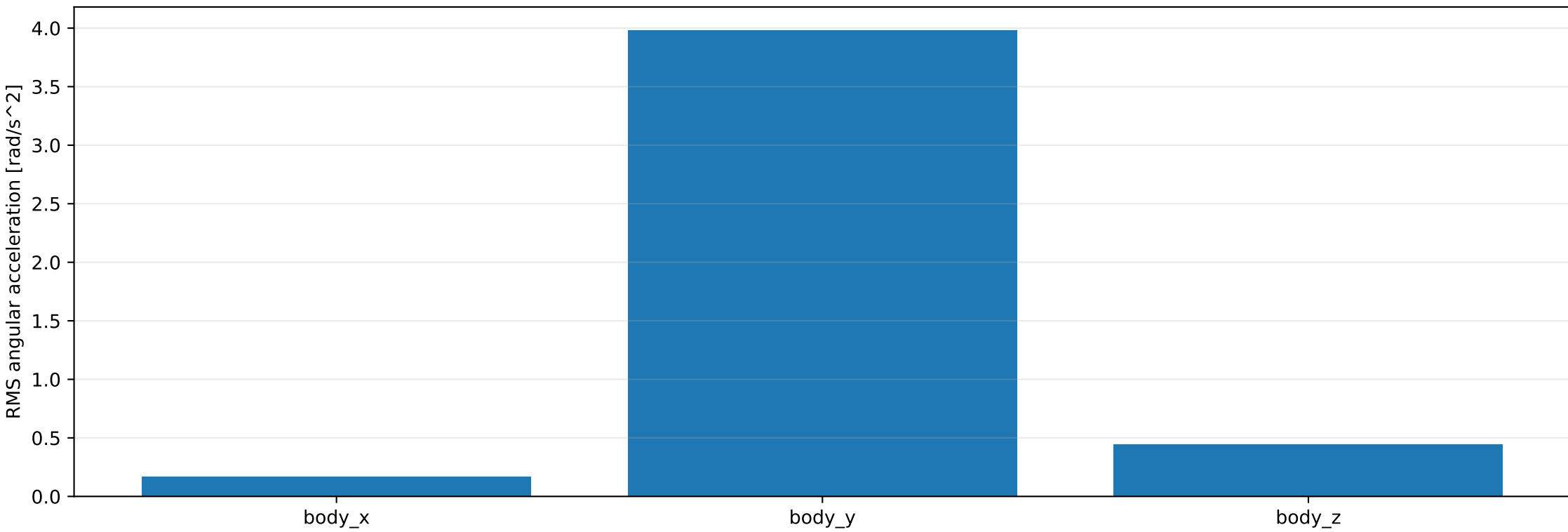
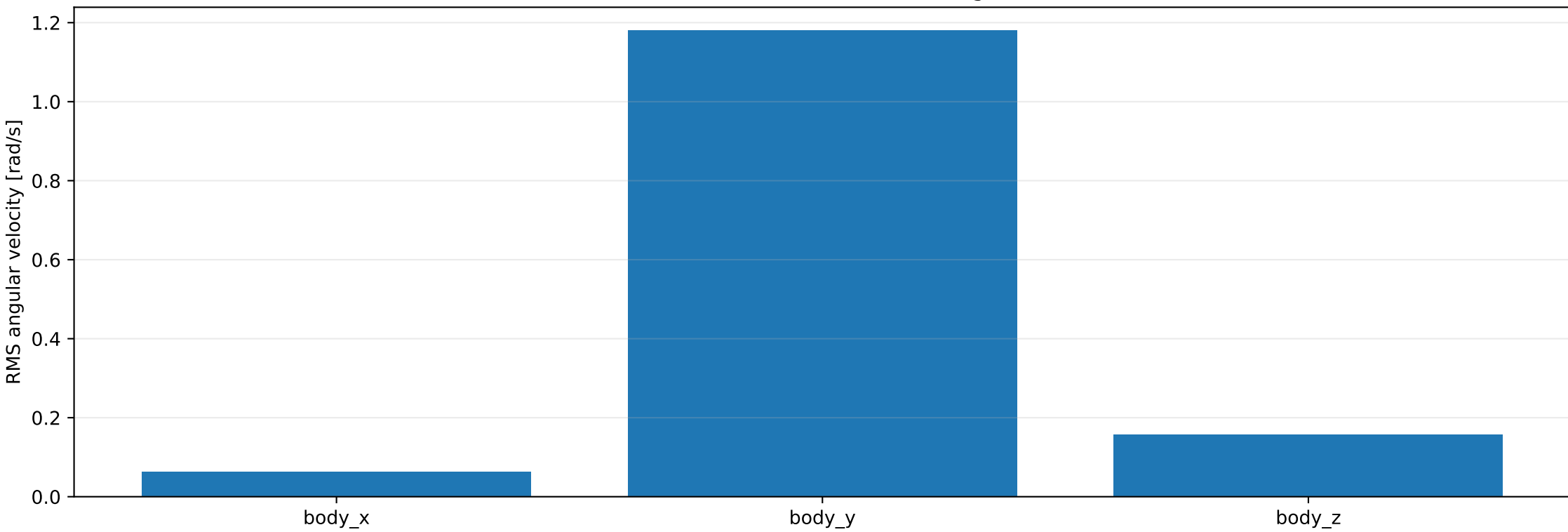
rotor force effectiveness

[0.9629222, 0.9749871, 0.96202821, 0.95174983]

selected delay [s]

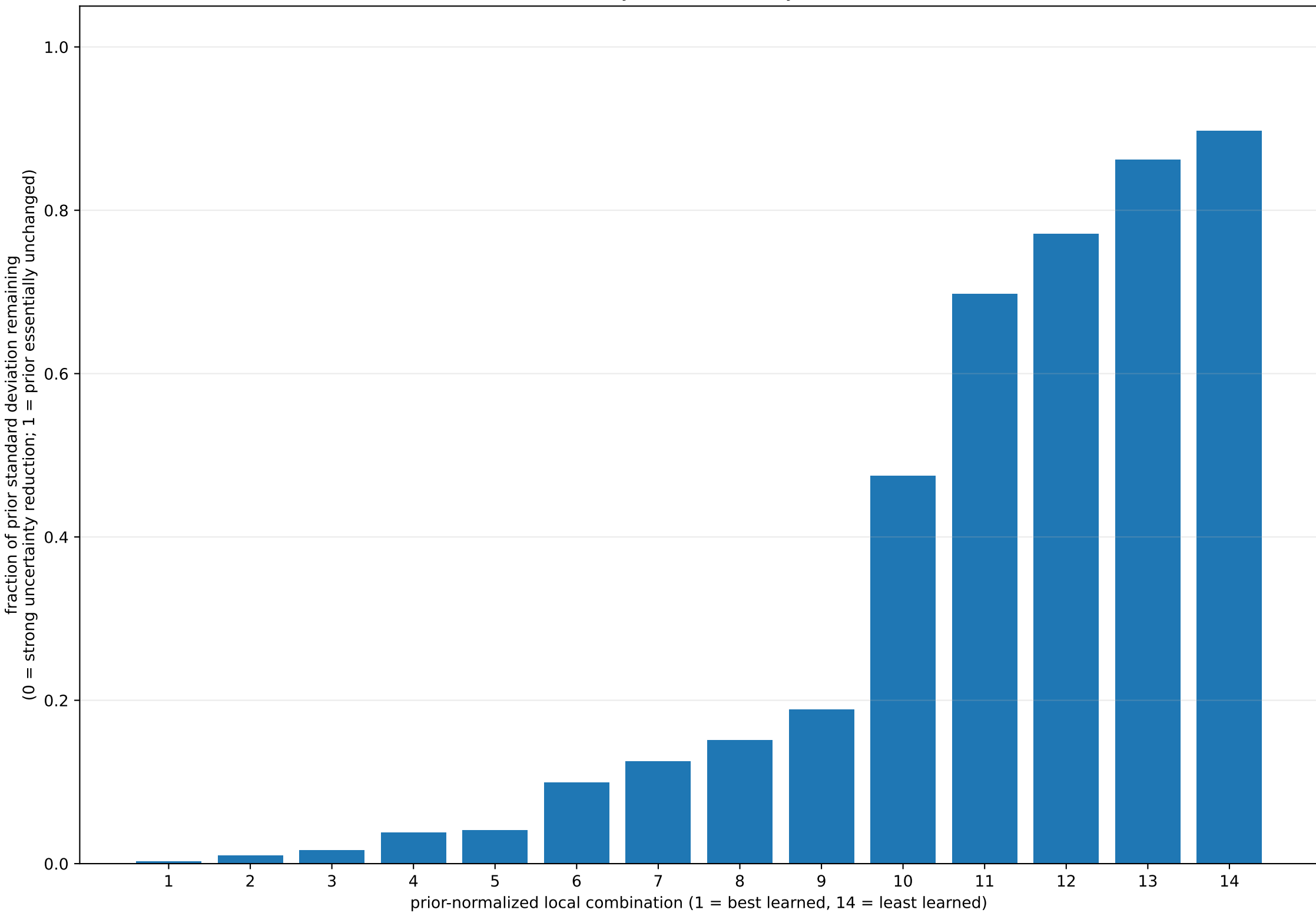
0.2

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

